

A fresh look at Australian stick insects reveals overlooked species

From: ScienceDaily Strange & Offbeat Mined: 2026-10-03 5 min read

Discovering a species does not always mean finding an animal nobody has seen before. Sometimes the animal has been photographed, collected and studied for years, but scientists have placed it in the wrong group. A new study of Australian stick insects shows how that can happen.

Researchers at the University of Sydney reexamined Australian members of *Anchiale*, a **genus**, or group of closely related species. Insects previously classified as just two species actually represent five. Together, they occur from northern New South Wales along coastal Queensland and into tropical northern Australia.

The team identified two previously unrecognized species and restored recognition of a third that had gone unrecognized for decades. The findings were published in *Austral Entomology*.

One particularly surprising case involved a large insect already familiar to scientists and insect enthusiasts. Evidence showed that it had been incorrectly identified for decades. It is now recognized as a separate species, *Anchiale robusta*.

To untangle these identities, the researchers combined DNA analysis with museum **specimens**, field surveys, citizen science observations and studies of insects raised in captivity. DNA provides inherited biological information, while **specimens** and observations let researchers compare the animals themselves. Using several kinds of evidence helped them examine distinctions that earlier **classifications** had missed.

There was already plenty of information to work with. More than 1,000 citizen science observations had been submitted through iNaturalist and the Atlas of Living Australia. Yet the study found overlooked **diversity** among insects that had also received decades of scientific attention.

Another newly described species, *Anchiale mabiensis*, comes from the critically endangered Mabi rainforest on Queensland's Atherton Tablelands. Its eggs have a distinctive structural lid. The researchers say this feature had not previously been documented in any stick insect or its closest relatives.

The revised **classification** could also help forestry and agriculture. One stick insect known for occasional large population outbreaks, during which insects strip leaves from trees, turned out to represent more than one species. Distinguishing those species could help researchers and land managers better understand and manage future impacts.

The team also identified a possible additional species from Cape York. Its identity remains unconfirmed because more **specimens** are needed.

Lead author Dr. Braxton Jones described **taxonomy**, the science of identifying and classifying organisms, as detective work. He said the findings show how much scientists still have to learn about insect **diversity**. He also emphasized protecting threatened **ecosystems** and documenting species before they disappear without being recognized.

Word Treasure

genus -- A group of closely related species.

classification -- The way scientists organize living things into groups.

specimens -- Collected examples of living things used for scientific study.

diversity -- The variety of different living things.

taxonomy -- The science of identifying and classifying organisms.

ecosystems -- Living things and their surroundings interacting together.

Background you need

A genus groups closely related species. Taxonomy is the science of identifying and classifying organisms.

Citizen science observations are observations shared by members of the public. This study used them alongside other evidence.

Break it down

Introduces how a familiar animal can have an overlooked scientific identity.

Reports the revised species count and publication.

Explains the recognition of *Anchiale robusta* and the research methods.

Describes existing public observations and the unusual eggs of *Anchiale mabiensis*.

Discusses possible benefits for forestry and agriculture and an unconfirmed species.

Closes with Dr. Braxton Jones's views on taxonomy and protecting ecosystems.

Why it matters

The study shows that even well-studied insects can include overlooked species. More accurate classification could also help researchers and land managers understand and manage effects on forestry and agriculture.

Test what you remember

1. What was the study's main finding about Australian Anchiale insects?
 - (A) Two recognized species belonged to five.
 - (B) Five recognized species belonged to two.
 - (C) Four recognized species belonged to three.
 - (D) Three recognized species belonged to four.
2. Which set of evidence did researchers use to examine the insects' identities?
 - (A) Tree measurements, rainfall records and star maps
 - (B) Satellite images, rock samples and ocean surveys
 - (C) Weather charts, soil samples and river surveys
 - (D) DNA analysis, museum specimens and field surveys
3. What information had the public already contributed?
 - (A) More than 1,000 rainforest weather forecasts
 - (B) More than 1,000 citizen science observations
 - (C) More than 1,000 agricultural planning reports
 - (D) More than 1,000 museum building proposals
4. Where does the newly described Anchiale mabiensis come from?
 - (A) The coastal forests of northern New South Wales
 - (B) The mountain forests around the city of Sydney
 - (C) The tropical forests of the Cape York region
 - (D) The Mabi rainforest on the Atherton Tablelands
5. Why does the possible additional species from Cape York remain unconfirmed?
 - (A) Researchers still need additional specimens.
 - (B) Researchers still need new rainfall records.
 - (C) Researchers still need museum floor plans.
 - (D) Researchers still need forestry proposals.
6. What did Dr. Braxton Jones compare taxonomy to?
 - (A) Building a shelter
 - (B) Predicting the weather
 - (C) Solving a detective case
 - (D) Planning a journey

Answer key (try the quiz first!): 1: A 2: D 3: B 4: D 5: A 6: C

Questions to think about

1. Dr. Braxton Jones: He compared taxonomy to detective work and said the discoveries show how much remains to be learned about insect diversity. He emphasized protecting threatened ecosystems and documenting species before they disappear without being recognized.

MY THOUGHTS (at least 20 words):